MS&E 228: Applied Causal Inference Powered by ML and AI

Lecture 16: Proximal Causal Inference

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Readings: *Applied Causal Inference Powered by ML and AI*. Tchetgen Tchetgen, Cui, Miao (2020). Cui et al. (2024).

Goals of this Lecture

Dealing with unobserved confounding: beyond sensitivity analysis and IV.

1. **Incomplete Proxies:** Can we use incomplete proxies for an unobserved confounder to get a *point estimate* of the causal effect?
2. **Connection to IV:** How does proximal causal inference relate to instrumental variables?
3. **Estimation:** How do we estimate proximal causal effects using DML-style cross-fitting?
4. **Broad Applicability:** Where has proximal inference been applied in practice, and what are the frontiers?

Part I: Recap & Activity

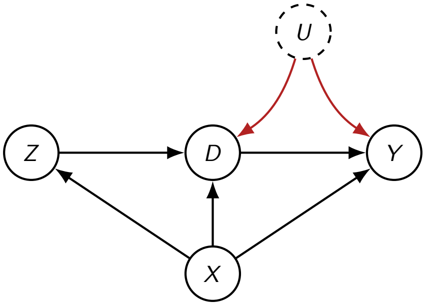
Instrumental Variables from Last Lecture

Quick Recap: The IV Approach

We introduced **Instrumental Variables** as a strategy for unobserved confounding.

Key idea: Find variable Z (instrument) that:

1. Affects the treatment D (*relevance*).
2. Has no direct effect on Y except through D (*exclusion restriction*).
3. Is as-if random, conditional on X (*exogeneity*).



For binary treatment / instrument:

$$\text{LATE} = \frac{\text{ATE}(Z \to Y)}{\text{ATE}(Z \to D)}$$

Under Partial Linearity: The IV *moment* condition:

$$\mathbb{E}[(\widetilde{Y} - \theta\widetilde{D}) \cdot \widetilde{Z}] = 0 \implies \text{ATE} := \theta = \frac{\text{Cov}(\widetilde{Z}, \widetilde{Y})}{\text{Cov}(\widetilde{Z}, \widetilde{D})}$$

Activity: Spot the Instrument

Health Insurance and Emergency Room Visits

A state government wants to know whether expanding Medicaid coverage reduces emergency room (ER) visits. People who sign up for Medicaid may be systematically sicker or poorer than those who do not. In Oregon in 2008, the state expanded Medicaid through a lottery: eligible residents could enter a lottery, and winners were given the opportunity to apply for Medicaid.

Questions for the group:

- What is the instrument (Z), treatment (D), and outcome (Y)?
- State the exclusion restriction. Is it plausible?
- Who are the compliers?

Activity: Spot the Instrument

Online Recommendations and User Engagement

A travel website wants to estimate the causal effect of users becoming premium members on the number of days they visit the site. Users who choose to become premium members are likely already more engaged. The company runs an A/B test: a random subset of users sees an "easier signup" flow (fewer clicks, prominent button), while the control group sees the standard flow.

Questions for the group:

- What is the instrument (Z), treatment (D), and outcome (Y)?
- State the exclusion restriction. Is it plausible?
- Who are the compliers?

Part II: Proximal Causal Inference

Leveraging Incomplete Proxies for Unobserved Confounders

Complete vs. Incomplete Proxies

In past lectures, we discussed looking for proxies of an unobserved confounder U:

- **Complete proxy:** A variable that is a *complete mediator* on the path from U to D (a complete treatment proxy) or from U to Y (a complete outcome proxy). Controlling for one such proxy suffices.
- **Incomplete proxy:** A variable influenced by U, but U still has other paths of influence. Controlling for it *reduces* but does not *eliminate* the omitted variable bias.
- **Sensitivity analysis:** We can bound the remaining bias, and the stronger the proxy, the tighter the bounds.

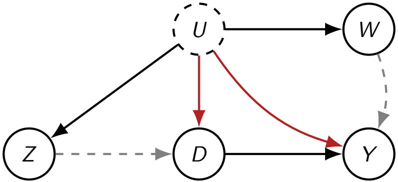
The Question

Is there an identification strategy that can leverage *incomplete* proxies to get a **point estimate**—not just bounds?

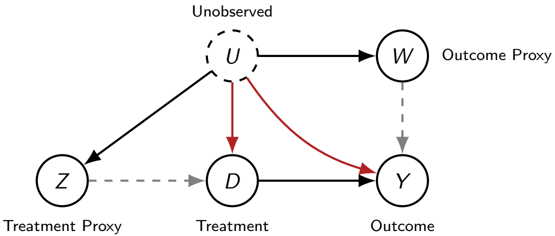
The Key Insight: Two Types of Proxies

Yes! If we can find two types of incomplete proxies, we can achieve point identification.

1. **Treatment proxy** Z ("negative control treatment"): a variable influenced by U that is (potentially) related to D, but has *no direct effect on* Y once we condition on D and U.
2. **Outcome proxy** W ("negative control outcome"): a variable influenced by U that is (potentially) related to Y, but the *treatment* D *has no causal effect on* W once we condition on U.



The Proximal Causal Inference DAG

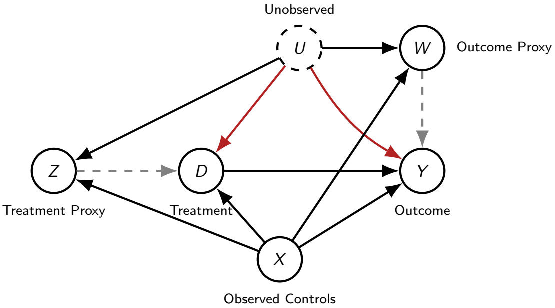


Three Crucial Assumptions

1. $Y \perp\!\!\!\perp Z \mid D, U$ *(Z has no direct effect on Y)*
2. $W \perp\!\!\!\perp D \mid U$ *(D has no causal effect on W)*
3. $Y(d) \perp\!\!\!\perp D \mid U$ *(conditional exogeneity if U were observed)*

The Proximal Causal Inference DAG

The full structure, including observed controls X:



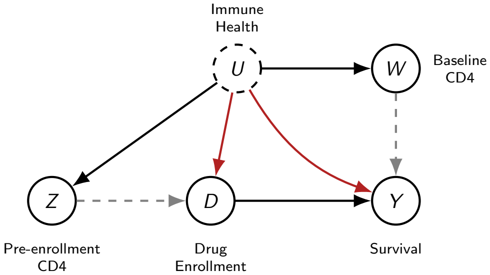
Three Crucial Assumptions

1. $Y \perp\!\!\!\perp Z \mid D, U, X$ — *(Z has no direct effect on Y)*
2. $W \perp\!\!\!\perp D \mid U, X$ — *(D has no causal effect on W)*
3. $Y(d) \perp\!\!\!\perp D \mid U, X$ — *(conditional exogeneity if U were observed)*

Motivating Example: HIV Drug Effectiveness

We want to estimate the effect of enrolling in a novel HIV drug therapy (D) on survival (Y).

- **Unobserved confounder U:** The patient's overall immune health (as it pertains to HIV).
- **Treatment proxy Z:** Pre-enrollment CD4 cell count (lab measurements taken before the drug decision). Doctors use these *plus their own clinical assessment* to decide on treatment.
- **Outcome proxy W:** Baseline post-enrollment CD4 cell count (measured right after enrollment, before the drug can have any effect).



Even if we control for Z, the physician's clinical assessment creates another confounding path $Y \leftarrow U \rightarrow D$ that is not blocked.

Why "Negative Controls"?

Proxies are called "negative controls" because of how they were *traditionally* used:

Traditional Use: Checking for Confounding

- **Negative control treatment Z:** If there were no unobserved confounding, Z should have *no association* with Y once we condition on D and observed controls X. Finding an association is evidence of confounding.
- **Negative control outcome W:** If there were no unobserved confounding, D should have *no association* with W, conditional on observed controls X. Finding an association is evidence of confounding.

The Breakthrough: Correcting for Confounding

If you have *both* types of negative controls simultaneously, you can go beyond just detecting confounding—you can **correct for it** and obtain a point estimate of the causal effect!

Poll Question

Poll: Why Not Just Control for Z?

In the HIV drug example, why can't we just control for the pre-enrollment CD4 count Z as a regular covariate to remove confounding?

1. Because Z is a collider between D and Y.
2. Because Z is an incomplete proxy—controlling for it reduces but does not eliminate the bias from U.
3. Because Z is a mediator on the causal path from D to Y.
4. Because Z is not measured in the data.

Empirical Applications of Proximal Causal Inference

Domain	**Treatment**	**Proxy** Z	**Proxy** W	**Unobs.** U
Vaccine Effectiveness	Flu Vaccine	Prior-season vaccination	Off-season illness	Health-seeking behavior
Cardiovascular Effects	Flu Vaccine	GERD visit	Cataract visit	Frailty
Peer Effects	Peer GPA	Peer-of-peer GPA	Peer headaches	Homophily
Text as Proxy	Vasopressor use	ECG note text	Nursing note text	Disease severity
Implicit Bias in Clinical Decisions	Treatment decisions	Pre-treatment labs	Baseline labs	Patient health status

Sources: Shi et al. (2020), Guo et al. (2025), Egami & Tchetgen Tchetgen (2024), Chen et al. (2024), Li et al. (2023).

Part III: Identification in the Linear Model

How Proximal Inference Works

The Linear Structural Equations

Let us work through the mechanics in a simple linear model. Suppose all variables are scalar and de-meaned (for now, no further observed controls X):

$$
\begin{aligned}
Y &= \theta D + \beta W + \gamma U + \varepsilon_Y \\
W &= \delta U + \varepsilon_W \\
D &= f_D(Z, U, \varepsilon_D) \\
Z &= f_Z(U, \varepsilon_Z)
\end{aligned}
$$

- θ is the **causal effect** of D on Y that we want to identify.
- U is unobserved. The functions f_D, f_Z can be arbitrary.
- The outcome equation for Y and the equation for W are **linear** in U.

The Substitution Trick

Key observation: We can replace the unobserved U in the outcome equation using the equation for W.

From $W = \delta U + \varepsilon_W$, we get $U = \frac{1}{\delta}(W - \varepsilon_W)$. Substituting:

$$\begin{aligned} Y &= \theta D + \beta W + \frac{\gamma}{\delta}(W - \varepsilon_W) + \varepsilon_Y \\ &= \theta D + \underbrace{\left(\beta + \frac{\gamma}{\delta}\right)}_{\kappa} W + \underbrace{\varepsilon_Y - \frac{\gamma}{\delta}\varepsilon_W}_{\text{residual}} \end{aligned}$$

Rewritten Outcome Equation

$$Y = \theta D + \kappa W + \eta, \qquad \text{where } \eta = \varepsilon_Y - \frac{\gamma}{\delta}\varepsilon_W$$

This looks like a regression of Y on (D, W), but we **cannot** run OLS because W is correlated with η (due to ϵ_W).

Why the Residual is Uncorrelated with Z

Now the magic happens. The residual $\eta = \varepsilon_Y - \frac{\gamma}{\delta}\varepsilon_W$ is uncorrelated with Z!

- $\mathbb{E}[\varepsilon_Y \cdot Z] = 0$: ε_Y is a noise term that is "instantiated" after Z.
  This step is essentially because of the first assumption $Y \perp\!\!\!\perp Z \mid D, U$. Once we account for D and U, the treatment proxy Z has no remaining influence on Y.
- $\mathbb{E}[\varepsilon_W \cdot Z] = 0$: ε_W is a noise term that is instantiated after Z.
  This is essentially because of the second assumption $W \perp\!\!\!\perp D \mid U$. Once we account for U, the treatment has no influence on W.

Therefore:

$$\mathbb{E}[(Y - \theta D - \kappa W) \cdot Z] = \mathbb{E}[\eta \cdot Z] = 0.$$

Why the Residual is Uncorrelated with D

Moreover, the residual is also uncorrelated with D.

- $\mathbb{E}[\varepsilon_Y \cdot D] = 0$: ε_Y is a noise term that is "instantiated" after D.
- $\mathbb{E}[\varepsilon_W \cdot D] = 0$: ε_W is a noise term that is "instantiated" after D.

Therefore,

$$\mathbb{E}[(Y - \theta D - \kappa W) \cdot D] = \mathbb{E}[\eta \cdot D] = 0$$

The reason why could claim this property is essentially due to the third crucial assumption, conditional exogeneity given U.

The Moment Equations

Combining, we have two moment conditions and two unknowns (θ, κ):

$$\mathbb{E}\left[(Y - \theta D - \kappa W)\begin{pmatrix} D \\ Z \end{pmatrix}\right] = \mathbf{0}$$

In matrix form:

$$\underbrace{\mathbb{E}\left[\begin{pmatrix} D \\ Z \end{pmatrix}\begin{pmatrix} D & W \end{pmatrix}\right]}_{\mathrm{Cov}\left(\binom{D}{Z},\binom{D}{W}\right)}\begin{pmatrix} \theta \\ \kappa \end{pmatrix} = \underbrace{\mathbb{E}\left[\begin{pmatrix} D \\ Z \end{pmatrix} Y\right]}_{\mathrm{Cov}\left(\binom{D}{Z},Y\right)}$$

Identification Condition

System has a unique solution if covariance matrix $\mathrm{Cov}\left(\binom{D}{Z}, \binom{D}{W}\right)$ is **full rank**.

The coefficient θ on D is exactly the **causal effect** we care about!

The Connection to Instrumental Variables

Key Insight: Proximal Inference is a Generalized IV Problem

Recall the IV moment condition from last lecture:

$$\mathbb{E}[(Y - \theta D) \cdot Z] = 0$$

Find θ so the residual is uncorrelated with the instrument Z.

In proximal inference, we have:

$$\mathbb{E}\left[(Y - \theta D - \kappa W) \cdot \begin{pmatrix} D \\ Z \end{pmatrix}\right] = \mathbf{0}$$

Find (θ, κ) so the residual is uncorrelated with the "instruments" (D, Z).

In essence, proximal inference is an IV method where:

- The "endogenous variables" (treatments) are $(\boldsymbol{D}, \boldsymbol{W})$.
- The "instruments" are $(\boldsymbol{D}, \boldsymbol{Z})$.

Solution is a "Multivariate" IV Formula

Key Insight: Proximal Inference is a Generalized IV Problem

The solution to the IV moment condition is the ratio of two covariances:

$$\theta = \mathrm{Cov}(Z, D)^{-1}\mathrm{Cov}(Z, Y)$$

In proximal inference, we have "generalized ratio":

$$\begin{pmatrix}\theta \\ \kappa\end{pmatrix} = \mathrm{Cov}\left(\begin{pmatrix}D \\ Z\end{pmatrix}, \begin{pmatrix}D \\ W\end{pmatrix}\right)^{-1}\mathrm{Cov}\left(\begin{pmatrix}D \\ Z\end{pmatrix}, Y\right)$$

In essence, proximal inference is an IV method where:

- The "endogenous variables" (treatments) are $(\boldsymbol{D}, \boldsymbol{W})$.
- The "instruments" are $(\boldsymbol{D}, \boldsymbol{Z})$.

Part IV: DML for Proximal Inference

Cross-Fitting and Practical Estimation

Adding Covariates: The Partially Linear Model

In practice, we also have observed controls X that can affect all variables:

$$
\begin{aligned}
Y &= \theta D + \beta W + \gamma U + g_Y(X) + \varepsilon_Y \\
W &= \delta U + g_W(X) + \varepsilon_W
\end{aligned}
$$

Following the DML logic, we **residualize** all variables by subtracting their conditional expectation given X:

$$
\widetilde{Y} = Y - \mathbb{E}[Y|X], \quad \widetilde{D} = D - \mathbb{E}[D|X], \quad \widetilde{W} = W - \mathbb{E}[W|X], \quad \widetilde{Z} = Z - \mathbb{E}[Z|X]
$$

The residualized equation is $\widetilde{Y} = \theta\widetilde{D} + \kappa\widetilde{W} + \eta$. We are back to the case without X's, but on residuals. The same moment conditions hold:

$$
\mathbb{E}\left[(\widetilde{Y} - \theta\widetilde{D} - \kappa\widetilde{W})\begin{pmatrix}\widetilde{D}\\ \widetilde{Z}\end{pmatrix}\right] = \mathbf{0}
$$

The DML-Proximal Estimator

As with DML-IV / DML, the formula is **insensitive** to estimation errors in the residualization models. This leads to a simple algorithm:

1. Compute cross-fitted residuals $\check{Y}, \check{D}, \check{W}, \check{Z}$.
2. Solve the empirical moment condition:

DML-Proximal Estimator

$$\begin{pmatrix} \hat{\theta} \\ \hat{\kappa} \end{pmatrix} = \mathbb{E}_n\left[\begin{pmatrix} \check{D} \\ \check{Z} \end{pmatrix} \begin{pmatrix} \check{D} & \check{W} \end{pmatrix}\right]^{-1} \mathbb{E}_n\left[\begin{pmatrix} \check{D} \\ \check{Z} \end{pmatrix} \check{Y}\right]$$

Note: Here Z and W can be *vectors*. If $\dim(Z) = \dim(W)$, the system is exactly identified; if $\dim(Z) > \dim(W)$, it is overidentified (discussed next).

Influence Function and Asymptotic Variance

The estimator $(\hat{\theta}, \hat{\kappa})$ is **asymptotically normal**:

$$\sqrt{n}\begin{pmatrix}\hat{\theta}-\theta \\ \hat{\kappa}-\kappa\end{pmatrix} \xrightarrow{d} N\Big(\mathbf{0},\ \mathbb{E}[\rho_i\rho_i^\top]\Big)$$

where the **influence function** ρ_i for observation i is:

Influence Function

$$\rho_i = \underbrace{\mathbb{E}\left[\begin{pmatrix}\widetilde{D} \\ \widetilde{Z}\end{pmatrix}\begin{pmatrix}\widetilde{D} & \widetilde{W}\end{pmatrix}\right]^{-1}}_{\text{Jacobian inverse}} \underbrace{\begin{pmatrix}\widetilde{D}_i \\ \widetilde{Z}_i\end{pmatrix}}_{\text{instruments}} \underbrace{\left(\widetilde{Y}_i - \theta\widetilde{D}_i - \kappa\widetilde{W}_i\right)}_{\text{residual}}$$

This is a **generalized IV sandwich formula**. If $Z = W$ (i.e. instruments = regressors), this reduces to the standard OLS sandwich variance.

Python Pseudocode: DML-Proximal

```python
def dml_proximal(X, Z, W, D, Y, models, nfolds=5):
  # X:(n,p_x), Z:(n,p_z), W:(n,p_w), D:(n,), Y:(n,)
  n = X.shape[0]
  kf = KFold(n_splits=nfolds, shuffle=True)  # same folds for all
  res_y, res_d = np.zeros(n), np.zeros(n)
  res_w, res_z = np.zeros((n, W.shape[1])), np.zeros((n, Z.shape[1]))
  # Step 1: Cross-fit residuals (coupled folds across all models)
  for train_idx, test_idx in kf.split(X):
    for var, res, key in [(Y,res_y,'y'),(D,res_d,'d'),
                          (W,res_w,'w'),(Z,res_z,'z')]:
      m = models[key].fit(X[train_idx], var[train_idx])
      res[test_idx] = var[test_idx] - m.predict(X[test_idx])
  # Step 2: Solve moment equations E[(D;Z)(D,W)'] coef = E[(D;Z) Y]
  R = np.column_stack([res_d, res_z])  # instruments (n, 1+p_z)
  T = np.column_stack([res_d, res_w])  # treatments  (n, 1+p_w)
  A = R.T @ T / n;   b = R.T @ res_y / n
  coef = np.linalg.lstsq(A, b, rcond=None)[0]  # (1+p_w,)
  theta_hat = coef[0]  # causal effect of D
  # Step 3: Influence function & sandwich standard errors
  eps = res_y - T @ coef
  Ainv_R = np.linalg.lstsq(A, R.T, rcond=None)[0]  # A^{-1} R'
  infl = Ainv_R * eps[np.newaxis, :]  # (1+p_w) x n
  V = (infl @ infl.T) / n**2
  se_theta = np.sqrt(V[0, 0])
  return theta_hat, se_theta
```

Handling More Instruments than Proxies

What if we have more treatment proxies Z than outcome proxies W?

For instance, multiple pre-enrollment lab measurements $Z_1, \ldots, Z_p$ but only one baseline outcome proxy W.

- We have **more moment equations than unknowns**.
- Since the residual is uncorrelated with all Z's, it is also uncorrelated with any *linear combination* of them.

Solution: Project $\widetilde{W}$ onto $\widetilde{Z}$

Find the linear combination of $\widetilde{Z}$'s that best predicts $\widetilde{W}$:

$$B^* = \arg\min_B \mathbb{E}\left[\|\widetilde{W} - B\widetilde{Z}\|^2\right]$$

Then use $\bar{Z} = B^*\widetilde{Z}$ as the instrument instead of the full $\widetilde{Z}$.

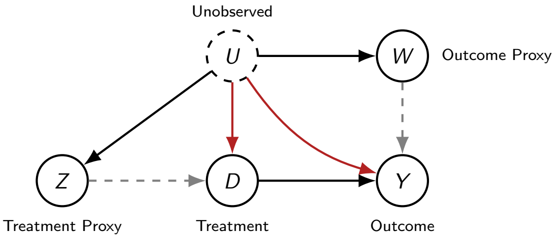
Full DML-Proximal with Many Instruments

```
def dml_proximal_overid(X, Z, W, D, Y, models, nfolds=5):
  # Z:(n,p_z), W:(n,p_w) with p_z > p_w (overidentified)
  n, p_w = X.shape[0], W.shape[1]
  kf = KFold(n_splits=nfolds, shuffle=True)  # same folds for all
  res_y, res_d = np.zeros(n), np.zeros(n)
  res_w, res_z = np.zeros((n,p_w)), np.zeros((n,Z.shape[1]))
  z_bar = np.zeros((n, p_w))  # projected instruments
  trained_models = {}
  for train_idx, test_idx in kf.split(X):
    # Step 1: Residualize all variables (coupled folds)
    for var, res, key in [(Y,res_y,'y'), (D,res_d,'d'), (W,res_w,'w'), (Z,res_z,'z')]:
      m = models[key].fit(X[train_idx], var[train_idx])
      res[test_idx] = var[test_idx] - m.predict(X[test_idx])
      trained_models[key] = m
    # Step 2: Out-of-fold projection W ~ Z (learn B on train)
    res_z_train = Z[train_idx] - trained_models['z'].predict(X[train_idx])
    res_w_train = W[train_idx] - trained_models['w'].predict(X[train_idx])
    B = OLS().fit(res_z_train, res_w_train)
    z_bar[test_idx] = B.predict(res_z[test_idx])  # apply on test
  # Step 3: Solve moment equations using z_bar as instruments
  R = np.column_stack([res_d, z_bar])  # (n, 1+p_w)
  T = np.column_stack([res_d, res_w])  # (n, 1+p_w)
  A = R.T @ T / n;  b = R.T @ res_y / n
  coef = np.linalg.solve(A, b);  theta_hat = coef[0]
  # Step 4: Influence function & sandwich standard errors
  eps = res_y - T @ coef
  Ainv_R = np.linalg.inv(A) @ R.T  # (1+p_w) x n
  infl = Ainv_R * eps[np.newaxis, :]
  V = (infl @ infl.T) / n**2
  se_theta = np.sqrt(V[0, 0])
  return theta_hat, se_theta
```

Part V: Beyond Partial Linearity

The Non-Parametric Proximal Problem

The Proximal Causal Inference DAG



Three Crucial Assumptions

1. $Y \perp\!\!\!\perp Z \mid D, U$ *(Z has no direct effect on Y)*
2. $W \perp\!\!\!\perp D \mid U$ *(D has no causal effect on W)*
3. $Y(d) \perp\!\!\!\perp D \mid U$ *(conditional exogeneity if U were observed)*

The Non-Parametric Proximal Problem

In the linear case, we looked for a *linear model of* D, W that would make the residuals *un-correlated with* D, Z.

Non-Parametric Moment Condition

Now we look for a flexible predictive model $h(D, W)$ of Y that would make the residuals independent of D, Z:

$$\mathbb{E}[Y - h(D, W) \mid D, Z] = 0$$

The Non-Parametric Proximal Problem

In the linear case, the linear model of D, W, was not a causal model of D, W on Y. Parameter κ was not effect of W on Y, only θ was correct effect of D on Y.

Proximal g-formula

We ignore model's h behavior in terms of W and only look at the "effect" it produces when we flip the treatment variable on / off, averaging over W:

$$\text{ATE} = \mathbb{E}[h(1, W) - h(0, W)]$$

This is a **g-formula**, where h is our predictive model of Y.

If h was a linear model $\theta D + \kappa W$, this would simply return θ.

The Non-Parametric Proximal Problem

Non-Parametric Moment Condition

Now we look for a flexible model $h(D, W)$ that would make the residuals independent of D, Z:

$$\mathbb{E}[Y - h(D, W) \mid D, Z] = 0$$

The average treatment effect is recovered via a **g-formula** analogue:

$$\text{ATE} = \mathbb{E}[h(1, W) - h(0, W)]$$

Key Extra Assumption

The non-parametric moment condition problem admits a solution h. And the set of solutions is the same as the set of solutions to the problem (completeness):

$$\mathbb{E}[Y - h(D, W) \mid D, U] = 0$$

The Non-Parametric Conditional Moment Problem

Non-Parametric Moment Condition

Now we look for a flexible model $h(D, W)$ that would make the residuals independent of D, Z:

$$\mathbb{E}[Y - h(D, W) \mid D, Z] = 0$$

- The function $h(D, W)$ is **not** a standard regression model of Y on (D, W).
- It is a model of (D, W) whose residual is mean-zero *conditional on a different set of variables* (D, Z).
- We are not looking for the best predictive model of Y using D, W.
- We cannot just minimize the expected squared loss.

Methods for the NPIV Problem

Solving for h such that $\mathbb{E}[Y - h(D, W) \mid D, Z] = 0$ is called the **Non-Parametric Instrumental Variable Regression** (NPIV) problem. Two main approaches:

1. Two-Stage Approach (DeepIV):

$$\min_h \mathbb{E}\left[\left(Y - \hat{\mathbb{E}}[h(D, W) \mid D, Z]\right)^2\right] + \lambda \mathbb{E}[h(D, W)^2]$$

Estimate the conditional density of (D, W) given (D, Z), then optimize h.

2. Adversarial / Minimax Approach:

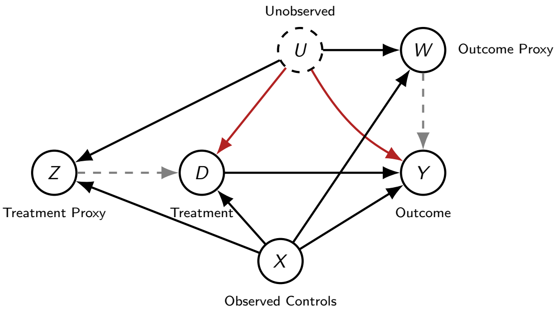
$$\min_h \max_f \mathbb{E}[(Y - h(D, W)) \cdot f(D, Z)] - \mathbb{E}[f(D, Z)^2] + \lambda \mathbb{E}[h(D, W)^2]$$

A "learner" chooses h; an "adversary" finds a critic f to expose violations. Solved via gradient descent/ascent, similar to training GANs.

These are active research areas with practical implementations, but more complex than the partially linear case.

The Proximal Causal Inference DAG

Same results extend with covariates.



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The Non-Parametric Proximal Problem

Non-Parametric Moment Condition

Now we look for a flexible model $h(D, W, X)$ that would make the residuals independent of D, Z, X:

$$\mathbb{E}[Y - h(D, W, X) \mid D, Z, X] = 0$$

The average treatment effect is recovered via a **g-formula** analogue:

$$\text{ATE} = \mathbb{E}[h(1, W, X) - h(0, W, X)]$$

Part VII: In-Class Activity & Takeaways

Front-Door vs. IV vs. Proximal

Activity: Comparing Identification Strategies

We have now seen three strategies for dealing with unobserved confounding. Let's recap.

Discuss with your neighbor (3 minutes): For each strategy, what is the *key structural feature* of the causal graph that makes it work?

Three Strategies for Unobserved Confounding

1. **Front-Door Criterion** (Lecture 14)
2. **Instrumental Variables** (Lecture 15)
3. **Proximal Causal Inference** (Today)

For each, think about:

- What special variable(s) do you need?
- What must that variable *not* do (exclusion-type restriction)?
- What must that variable *do* (relevance-type condition)?

Distinguishing the Three Strategies

	Front-Door	IV	Proximal
Special variable(s)	Mediator M	Instrument Z	Treatment proxy Z Outcome proxy W
Exclusion	D affects Y *only* through M; no direct $D \to Y$; No unobserved confounding between D and M or M and Y	Z affects Y *only* through D; no direct $Z \to Y$ No unobserved confounding between Z and Y.	$Z \perp\!\!\!\perp Y \mid D, U$; $W \perp\!\!\!\perp D \mid U$; $Y(d) \perp\!\!\!\perp D \mid U$
Relevance	No constraint	Z shifts D	Z, W each carry information about U
Identifies	ATE	LATE (binary D) or partial-linear effect	ATE (partial-linear or NPIV)

Recap: Matching Examples to Strategies

For each example we have seen in class, identify which strategy applies and *why*.

Example 1: Ride-Sharing and Tipping (Bellemare, Bloem & Wexler, 2024)

D = authorizing a shared ride, Y = tips, U = unobserved rider frugality.
The platform observes whether the ride was *actually shared* (M), which depends on the algorithm matching another rider on the same route.

Discuss: Which strategy? What is the key structural feature?

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Example 1: Ride-Sharing and Tipping (Bellemare, Bloem & Wexler, 2024)

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The platform observes whether the ride was *actually shared* (M), which depends on the algorithm matching another rider on the same route.

Discuss: Which strategy? What is the key structural feature?

Answer: Front-Door

M (actually sharing) is a **complete mediator**: D affects Y only through M. The algorithm's matching decision does not depend on the rider's unobserved frugality U, so M is unconfounded given D.

Recap: Matching Examples to Strategies

Example 2: Oregon Medicaid Lottery (Finkelstein et al., 2012)

$D =$ Medicaid coverage, $Y =$ ER visits, $U =$ unobserved health/socioeconomic status.
Oregon expanded Medicaid via a *lottery*: eligible residents entered a lottery, and winners could apply for coverage.

Discuss: Which strategy? What is the key structural feature?

Recap: Matching Examples to Strategies

Example 2: Oregon Medicaid Lottery (Finkelstein et al., 2012)

D = Medicaid coverage, Y = ER visits, U = unobserved health/socioeconomic status. Oregon expanded Medicaid via a *lottery*: eligible residents entered a lottery, and winners could apply for coverage.

Discuss: Which strategy? What is the key structural feature?

Answer: Instrumental Variables

The lottery win (Z) is an **instrument**: it is randomly assigned (exogenous), it shifts the probability of getting Medicaid (relevant), and it has no direct effect on ER visits except through coverage (exclusion). Identifies the LATE for *compliers*—those whose coverage status is changed by winning.

Recap: Matching Examples to Strategies

Example 3: HIV Drug Effectiveness (Tchetgen Tchetgen et al., 2020)

D = drug enrollment, Y = survival, U = patient's overall immune health.
Doctors use pre-enrollment lab measurements (Z) *plus their own clinical assessment* to decide on treatment. Baseline post-enrollment CD4 counts (W) are measured before the drug can have any effect.

Discuss: Which strategy? Why not IV?

Recap: Matching Examples to Strategies

Example 3: HIV Drug Effectiveness (Tchetgen Tchetgen et al., 2020)

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Discuss: Which strategy? Why not IV?

Answer: Proximal Causal Inference

Z (pre-enrollment CD4) is *not* an instrument—it does not satisfy the exclusion restriction because the physician's clinical assessment of U creates another path $U \to D$. Instead, Z is a **treatment proxy** and W (baseline CD4) is an **outcome proxy**. Together they enable proximal identification.

Why Not the Other Strategies?

A crucial part of choosing a strategy is understanding why the *other* strategies fail.

Example	**Correct Strategy**	**Why the others fail**
Ride-sharing tipping	Front-Door	No instrument available; no two distinct proxies for U
Oregon Medicaid	IV	No complete mediator; lottery is exogenous, not a proxy for U
HIV drug	Proximal	No complete mediator $D \to M \to Y$; pre-enrollment CD4 is not a valid instrument (physician sees U directly)

Key Takeaway

The choice of strategy is dictated by the **causal structure**—what special variables are available and what paths exist in the DAG. There is no universal "best" strategy; each requires a different structural feature.

Takeaways

1. **Negative controls detect AND correct:** Treatment proxies Z (negative control treatments) and outcome proxies W (negative control outcomes) were traditionally used to check for confounding. Having both types simultaneously enables **point identification** of the causal effect.
2. **Proximal inference generalizes IV:** In the partially linear setting, proximal causal inference reduces to an IV-like method where (D, W) are the "treatments" and (D, Z) are the "instruments," and the DML cross-fitting machinery applies directly.
3. **Practical and widely applied:** Proximal inference has been deployed in vaccine effectiveness, cardiovascular health, peer effects in education, clinical decision-making, and even with text data as proxies.
4. **Beyond linearity, NPIV methods apply:** In the fully non-parametric setting, proximal inference requires solving a Non-Parametric IV (NPIV) problem, for which modern ML methods (DeepIV, adversarial/minimax) provide practical solutions.

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